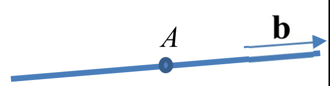
RAFFLES INSTITUTION
2026 Year 6 H2 Mathematics Timed Practice Revision Paper 2
Questions and Solutions with comments

Source: 2023 Y6CT Q1		
1	The points A, B and C have position vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ respectively. The points A and B are fixed while C varies.	
	(a)	Given that $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ are non-zero vectors such that $(\mathbf{c}-\mathbf{a})\times\mathbf{b}=\mathbf{b}\times(\mathbf{c}-\mathbf{a})$, find the relationship between $(\mathbf{c}-\mathbf{a})$ and $\mathbf{b}$. [2]
	(b)	Hence describe geometrically the set of all possible positions of the point C. [2]

Solution		
(a) **[2]**	Since $\mathbf{b}\times(\mathbf{c}-\mathbf{a})=-(\mathbf{c}-\mathbf{a})\times\mathbf{b}$, $(\mathbf{c}-\mathbf{a})\times\mathbf{b}=\mathbf{b}\times(\mathbf{c}-\mathbf{a})$ $\Rightarrow(\mathbf{c}-\mathbf{a})\times\mathbf{b}=-(\mathbf{c}-\mathbf{a})\times\mathbf{b}$ $\Rightarrow 2\left[(\mathbf{c}-\mathbf{a})\times\mathbf{b}\right]=\mathbf{0}$ $\Rightarrow(\mathbf{c}-\mathbf{a})\times\mathbf{b}=\mathbf{0}$ Hence, $(\mathbf{c}-\mathbf{a})$ and $\mathbf{b}$ are parallel or $\mathbf{c}=\mathbf{a}$.	$(\mathbf{c}-\mathbf{a})\times\mathbf{b}=\mathbf{0}$ implies $\lvert\mathbf{c}-\mathbf{a}\rvert\lvert\mathbf{b}\rvert\sin\theta=0$, **where** θ **is the angle between** $(\mathbf{c}-\mathbf{a})$ **and** $\mathbf{b}$**.** Thus either $\lvert\mathbf{c}-\mathbf{a}\rvert=0$ or $\sin\theta=0$ (Note $\mathbf{b}\neq\mathbf{0}$)
(b) **[2]**	From (a), since $\mathbf{c}=\mathbf{a}$ or $(\mathbf{c}-\mathbf{a})$ and $\mathbf{b}$ are parallel. $(\mathbf{c}-\mathbf{a})=k\mathbf{b},\ k\in\mathbb{R}$ $\mathbf{c}=\mathbf{a}+k\mathbf{b},\ k\in\mathbb{R}$  Point C lies on the line passing through the point A and the line is parallel to the vector $\mathbf{b}$. Note that $\mathbf{c}=\mathbf{a}$ when $k=0$.	

Source: 2021 Y6TP Q1		
2	**(i)**	Use the substitution $t=\sqrt{x}$ to show that $\displaystyle\int \frac{1}{\sqrt{x}\left(1+x\right)}\,\mathrm{d}x = 2\tan^{-1}\sqrt{x}+c$, where c is an arbitrary constant. [2]
	(ii)	Hence, find the exact value of $\displaystyle\int_1^4 \frac{\ln(1+x)}{x^{\frac{3}{2}}}\,\mathrm{d}x$. [4]

Solution		**Comments**
(i) **[2]**	Given the substitution $t=\sqrt{x}$, we have $\dfrac{\mathrm{d}t}{\mathrm{d}x}=\dfrac{1}{2\sqrt{x}}$ $\displaystyle\int \frac{1}{\sqrt{x}\left(1+x\right)}\,\mathrm{d}x$ $\displaystyle=2\int \frac{1}{\left(1+\left(\sqrt{x}\right)^2\right)}\left(\frac{1}{2\sqrt{x}}\right)\mathrm{d}x$ $\displaystyle=2\int \frac{1}{1+t^2}\,\mathrm{d}t = 2\tan^{-1}t+c$ $=2\tan^{-1}\sqrt{x}+c$	
(ii) **[4]**	$u=\ln(1+x)$ $\dfrac{\mathrm{d}v}{\mathrm{d}x}=x^{-\frac{3}{2}}$ $\dfrac{\mathrm{d}u}{\mathrm{d}x}=\dfrac{1}{1+x}$ $v=-2x^{-\frac{1}{2}}$ $\displaystyle\int_1^4 \frac{\ln(1+x)}{x^{\frac{3}{2}}}\,\mathrm{d}x$ $\displaystyle=\left[\left(-2x^{-\frac{1}{2}}\right)\ln(1+x)\right]_1^4-\int_1^4\frac{-2x^{-\frac{1}{2}}}{1+x}\,\mathrm{d}x$ $\displaystyle=\left[\left(-2x^{-\frac{1}{2}}\right)\ln(1+x)\right]_1^4+2\int_1^4\frac{1}{\sqrt{x}\left(1+x\right)}\,\mathrm{d}x$ $\displaystyle=\left[\left(-2x^{-\frac{1}{2}}\right)\ln(1+x)+4\tan^{-1}\sqrt{x}\right]_1^4$ $=-\ln 5+2\ln 2+4\tan^{-1}2-\pi$ $\left(\text{or } \ln\dfrac{4}{5}+4\tan^{-1}2-\pi\right)$	The application of part (i) does not happen directly, but observation of derivative of $\ln(1+x)$ and integration of $x^{-\frac{3}{2}}$ would lead to a term linked to part (i), suggesting integration by parts. Students are reminded to be careful of the negative sign in $v=-2x^{-\frac{1}{2}}$.

Source: 2019 Y5 CT Q3

3 Points O, A and B are such that $\overrightarrow{OA} = 2\mathbf{i} - \mathbf{k}$ and $\overrightarrow{OB} = 4\mathbf{i} + 5\mathbf{j} - 4\mathbf{k}$. The point C on OA produced is such that $OA : AC = 1 : 2$, and the point P on BC is such that $BP : PC = \lambda : 1-\lambda$.

(a) Show that $\overrightarrow{OP} = (4+2\lambda)\mathbf{i} + 5(1-\lambda)\mathbf{j} + (-4+\lambda)\mathbf{k}$. [2]

Given that OP is perpendicular to BC,

(b) find the value of λ, [3]

(c) find $\dfrac{\text{area of } \Delta OCP}{\text{area of } \Delta OAB}$. [2]

Solution

(i) [2]

$$\overrightarrow{OC} = 3\overrightarrow{OA} = 3\begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 6 \\ 0 \\ -3 \end{pmatrix}$$

By ratio theorem,

$\overrightarrow{OP}$

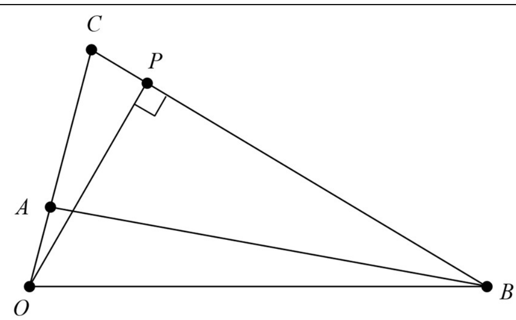
$$= (1-\lambda)\overrightarrow{OB} + \lambda\overrightarrow{OC}$$

$$= (1-\lambda)\begin{pmatrix} 4 \\ 5 \\ -4 \end{pmatrix} + \lambda\begin{pmatrix} 6 \\ 0 \\ -3 \end{pmatrix}$$

$$= \begin{pmatrix} 4(1-\lambda)+6\lambda \\ 5(1-\lambda) \\ -4(1-\lambda)-3\lambda \end{pmatrix}$$

$$= (4+2\lambda)\mathbf{i} + 5(1-\lambda)\mathbf{j} + (-4+\lambda)\mathbf{k} \quad \text{(shown)}$$

(ii) [3]



$\overrightarrow{BC} = 2\mathbf{i} - 5\mathbf{j} + \mathbf{k}$

Since OP is perpendicular to BC,

$\overrightarrow{OP} \cdot \overrightarrow{BC} = 0$

$$\begin{pmatrix} 4+2\lambda \\ 5-5\lambda \\ -4+\lambda \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -5 \\ 1 \end{pmatrix} = 0$$

$$8 + 4\lambda - 25 + 25\lambda - 4 + \lambda = 0$$

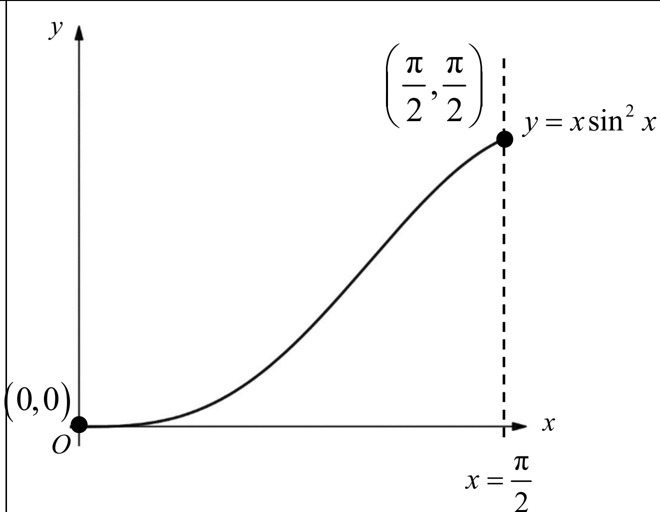
$$30\lambda = 21$$

$$\lambda = \frac{7}{10}$$

(iii) **[2]**	From (ii) $BP:PC=7:3$ So $BC:PC=10:3$ $\dfrac{\text{area of }\Delta OCP}{\text{area of }\Delta OCB}=\dfrac{\frac{1}{2}PC\times OP}{\frac{1}{2}BC\times OP}=\dfrac{3}{10}$ Similarly, ΔOAB and ΔOCB have the same height so, $\dfrac{\text{area of }\Delta OAB}{\text{area of }\Delta OCB}=\dfrac{OA}{OC}=\dfrac{1}{3}$ $\therefore\dfrac{\text{area of }\Delta OCP}{\text{area of }\Delta OAB}=\dfrac{9}{10}$	Alternatively, Area of $\Delta OCP=\dfrac{1}{2}\left\lvert\overrightarrow{OC}\times\overrightarrow{OP}\right\rvert$ $=\dfrac{1}{2}\left\lvert\overrightarrow{OC}\times\left(\dfrac{7}{10}\overrightarrow{OC}+\dfrac{3}{10}\overrightarrow{OB}\right)\right\rvert$ $=\dfrac{1}{2}\left\lvert 3\overrightarrow{OA}\times\dfrac{3}{10}\overrightarrow{OB}\right\rvert=\dfrac{9}{10}\times\dfrac{1}{2}\left\lvert\overrightarrow{OA}\times\overrightarrow{OB}\right\rvert$ $=\dfrac{9}{10}\left(\text{Area of }\Delta OAB\right)$ $\therefore\dfrac{\text{area of }\Delta OCP}{\text{area of }\Delta OAB}=\dfrac{9}{10}$

Source: 2023 Y6 TP Q1			
4	**(a)**	**(i)**	Find $\int \sin^2 x\,\mathrm{d}x$. [2]
		(ii)	Hence or otherwise, show that $$\int x\sin^2 x\,\mathrm{d}x=\frac{1}{4}x^2-\frac{1}{4}x\sin 2x-\frac{1}{8}\cos 2x+c.$$ [3]
	(b)	**(i)**	Sketch the graph of $y=x\sin^2 x$ for $0\le x\le\frac{1}{2}\pi$, labelling the end points clearly. [2]
		(ii)	Hence, find the exact area of the region bounded by the graph of $y=x\sin^2 x$, the x-axis and the line $x=\frac{1}{2}\pi$. [3]

Solution		**Comments**
(ai) **[2]**	$\int\sin^2 x\,\mathrm{d}x=\int\left(\dfrac{1-\cos 2x}{2}\right)\mathrm{d}x$ $=\dfrac{1}{2}\int(1-\cos 2x)\,\mathrm{d}x$ $=\dfrac{1}{2}\left(x-\dfrac{\sin 2x}{2}\right)+c$ $=\dfrac{1}{2}x-\dfrac{\sin 2x}{4}+c$	

<table>
<tr>
<td>(aii)
[3]</td>
<td>$\int x\sin^2 x\,\mathrm{d}x = x\left(\frac{1}{2}x-\frac{1}{4}\sin 2x\right)-\int\left(\frac{1}{2}x-\frac{1}{4}\sin 2x\right)\mathrm{d}x$
$= x\left(\frac{1}{2}x-\frac{1}{4}\sin 2x\right)-\left[\frac{1}{2}\left(\frac{x^2}{2}\right)-\frac{1}{4}\left(\frac{-\cos 2x}{2}\right)\right]+c$
$=\frac{1}{2}x^2-\frac{1}{4}x\sin 2x-\frac{1}{4}x^2-\frac{1}{8}\cos 2x+c$
$=\frac{1}{4}x^2-\frac{1}{4}x\sin 2x-\frac{1}{8}\cos 2x+c$ (shown)</td>
<td>“Hence” method:
Integration by parts:
<table><tr><td>$u=x$</td><td>$\frac{\mathrm{d}v}{\mathrm{d}x}=\sin^2 x$</td></tr><tr><td>$\frac{\mathrm{d}u}{\mathrm{d}x}=1$</td><td>$v=\frac{1}{2}x-\frac{\sin 2x}{4}$</td></tr></table></td>
</tr>
<tr>
<td></td>
<td>Alternatively,
$\int x\sin^2 x\,\mathrm{d}x = \int x\left(\frac{1-\cos 2x}{2}\right)\mathrm{d}x$
$=\frac{x^2}{4}-\frac{1}{2}\int x\cos 2x\,\mathrm{d}x$
$=\frac{x^2}{4}-\frac{1}{2}\left[x\left(\frac{\sin 2x}{2}\right)-\int\left(\frac{\sin 2x}{2}\right)\mathrm{d}x\right]$
$=\frac{x^2}{4}-\frac{x\sin 2x}{4}+\frac{1}{4}\left(\frac{-\cos 2x}{2}\right)+c$
$=\frac{1}{4}x^2-\frac{1}{4}x\sin 2x-\frac{1}{8}\cos 2x+c$ (shown)</td>
<td></td>
</tr>
<tr>
<td>(bi)
[2]</td>
<td>
</td>
<td>Students needs to zoom to the appropriate scale to see the shape of the graph. Do observe the gradient as well.</td>
</tr>
</table>

(bii) **[3]**	Required area $=\int_0^{\frac{\pi}{2}} x\sin^2 x\,\mathrm{d}x$ $=\left[\frac{1}{4}x^2-\frac{1}{4}x\sin 2x-\frac{1}{8}\cos 2x\right]_0^{\frac{\pi}{2}}$ $=\left[\frac{1}{4}\left(\frac{\pi}{2}\right)^2-\frac{1}{4}\left(\frac{\pi}{2}\right)\sin\left(2\left(\frac{\pi}{2}\right)\right)-\frac{1}{8}\cos\left(2\left(\frac{\pi}{2}\right)\right)\right]$ $-\left[\frac{1}{4}(0)^2-\frac{1}{4}(0)\sin(2(0))-\frac{1}{8}\cos(2(0))\right]$ $=\frac{\pi^2}{16}-\frac{1}{8}(-1)+\frac{1}{8}$ $=\left(\frac{\pi^2}{16}+\frac{1}{4}\right)\text{units}^2$	Some students made careless mistakes with the signs.

Source: 2023 Y6 CT Q3		
5	Cadence, the pedal speed in cycling, is measured in pedal stroke revolutions per minute (RPM). For example, a cadence of 60 RPM means that one pedal makes a complete revolution 60 times in one minute. Every complete revolution of the pedal stroke of a stationary bicycle is equivalent to cycling an approximate distance of 0.006 km. Lisa signed up for a beginner spinning class. The workout routine requires her to start cycling at 30 RPM consistently for the first minute and progressively increase the cadence by 2.5 RPM for every subsequent minute.	
	(a)	At which minute will Lisa cycle at 120 RPM? [2]
	(b)	Find Lisa's average speed of cycling, in km/h, for the first 15 minutes. [3]
	Celina signed up for an intermediate spinning class. The workout routine requires her to start cycling at 25 RPM consistently for the first minute and increase the cadence by 15% for every subsequent minute.	
	(c)	At which minute does Celina's cadence first exceed 120 RPM? [2]
	(d)	If Lisa and Celina start their workout routine at the same time, at which minute does Celina's total distance cycled first exceed Lisa's total distance cycled? [3]

Solution	**Comments**

(a) **[2]**	AP: $a=30,\ d=2.5$ $120=30+(n-1)(2.5)$ $n=37$ Lisa will cycle at 120 RPM at the 37th minute.	
(b) **[3]**	$S_{15}=\frac{15}{2}\left[2(30)+(14)(2.5)\right]$ $=712.5$ revolutions Average speed $=712.5\times 0.006\div\left(\frac{15}{60}\right)$ $=17.1$ km/h <u>Alternative Method using mean PRM</u> Mean PRM for first 15 minutes $=\frac{1}{2}\left[2(30)+(14)(2.5)\right]=47.5$ Average speed $=47.5\times 0.006\times 60$ $=17.1$ km/h	
(c) **[2]**	GP: $a=25,\ r=1.15$ $25(1.15)^{n-1}>120$ $n-1>11.223$ $n>12.223$ Celina's cadence will first exceed 120 RPM at the 13th minute.	
(d) **[3]**	Let the time taken by Celina's total distance to first exceed Lisa's to be *n*. $\frac{25(1.15^n-1)}{1.15-1}\times 0.006>\frac{n}{2}\left(2(30)+2.5(n-1)\right)\times 0.006$ $\frac{25}{0.15}(1.15^n-1)-\frac{n}{2}(57.5+2.5n)>0$ Let $Y=\frac{25}{0.15}(1.15^n-1)-\frac{n}{2}(57.5+2.5n)$, By GC, $n>5.8627$ **<u>OR</u>** use the table of values: <table><tr><td>n</td><td>Y</td></tr><tr><td>5</td><td>$-6.44<0$</td></tr><tr><td>6</td><td>$1.34>0$</td></tr></table> Celina's total distance cycled first exceed Lisa's total distance cycled at the 6th minute.	Reminder: when using GC, you still need to provide working, such as the inequalities. Table of values can be used if the unknown to be solved is an integer and you need to show the change in values, in this case < 0 and > 0.

Source: 2023 Y6 TP Q5

6 John models the nitrate level, N mg/L, present in the aquarium at a time t days after setting up his fish tank. The model assumes that, at any time t, the rate of increase of nitrate level due to ammonia and bacteria in the tank is kN mg/L per day, for some positive constant k. The initial nitrate level is 10 mg/L and it is found to be 20 mg/L after 1 day.

(a) **(i)** Write down a differential equation involving N, t and k. [1]

(ii) Solve this differential equation to find an expression for N in terms of t and show that $k = \ln 2$. [3]

(iii) Find the nitrate level after 3 days. [1]

(b) Hornwort is an aquatic plant that can help to reduce nitrate levels in aquariums. John adds s stalks of hornwort to his fish tank after 3 days. He then models the rate of decrease of nitrate level by the hornwort, x days after it is added, by h mg/L per stalk per day. The rate of increase of nitrate level due to ammonia and bacteria in the tank remains at $N \ln 2$ mg/L per day.

(i) Explain why the differential equation in this case is now $\dfrac{dN}{dx} = N\ln 2 - sh$. [1]

(ii) Solve this differential equation, giving N in terms of x, s and h. [3]

(c) In the case where $h = 18.5$, find the least number of stalks of hornwort needed in order for the nitrate level to decrease to 0 mg/L at $x = 7$. [3]

	Solution	**Comments**
(ai) **[1]**	$\dfrac{dN}{dt} = kN$	
(aii) **[3]**	$\displaystyle\int \frac{1}{N}\,dN = \int k\ dt$ Since $N > 0$, $\ln N = kt + c$, where $c \in \mathbb{R}$ $N = e^{kt+c} = e^{kt}e^{c}$ $N = Ae^{kt}$, where $A = e^{c}$ When $t = 0, N = 10$. Hence $A = 10$. When $t = 1, N = 20$. Hence, $20 = 10e^{k}$ $k = \ln 2$ (shown) $\therefore N = 10e^{t\ln 2}$ or $N = 10(2^{t})$	Note that the modulus sign in $\ln\lvert N\rvert$ should be included if there is no mention of N being positive. Question states that we need to find an expression for N in terms of t.

(aiii) **[1]**	$N = 10\mathrm{e}^{t\ln 2}$ or $N = 10(2^t)$ When $t = 3$, $N = 10\mathrm{e}^{3\ln 2} = 80$ The nitrate level after 3 days is 80 mg/L.	
(bi) **[1]**	Since h is the rate of decrease of nitrate level by one stalk of hornwort per day, sh would be the rate of decrease of nitrate level by s stalks of hornwort in one day. Total rate of change = (rate of increase due to ammonia and bacteria) – (rate of decrease due to s stalks of hornwort) hence $\frac{\mathrm{d}N}{\mathrm{d}x} = N\ln 2 - sh$.	
(bii) **[3]**	$\frac{\mathrm{d}N}{\mathrm{d}x} = N\ln 2 - sh$ $\frac{1}{N\ln 2 - sh}\frac{\mathrm{d}N}{\mathrm{d}x} = 1$ Integrate w.r.t x, $\int \frac{1}{N\ln 2 - sh}\,\mathrm{d}N = \int 1\,\mathrm{d}x$ $\frac{1}{\ln 2}\ln\lvert N\ln 2 - sh\rvert = x + d'$, where $d' \in \mathbb{R}$ $\ln\lvert N\ln 2 - sh\rvert = x\ln 2 + d$, where $d = d'\ln 2$ $\lvert N\ln 2 - sh\rvert = \mathrm{e}^{x\ln 2 + d}$ $N\ln 2 - sh = \pm\mathrm{e}^{x\ln 2}\mathrm{e}^{d}$ $N\ln 2 - sh = B\mathrm{e}^{x\ln 2}$, where $B = \pm\mathrm{e}^{d}$ $N = \frac{B\mathrm{e}^{x\ln 2} + sh}{\ln 2}$ ------- (*) When $x = 0$, $N = 80$. $80 = \frac{B + sh}{\ln 2}$ $B = 80\ln 2 - sh$ Substituting $B = 80\ln 2 - sh$ into (*), $N = \frac{(80\ln 2 - sh)\mathrm{e}^{x\ln 2} + sh}{\ln 2}$ $N = \frac{(80\ln 2 - sh)2^{x} + sh}{\ln 2}$	Note that the modulus sign for $\ln\lvert N\ln 2 - sh\rvert$ is required. The initial condition found in part **a(iii)** should be used to find the value of the arbitrary constant, B.
(c) **[3]**	**Method 1:** Given $h = 18.5$. When $x = 7$, $N = 0$. $0 = \frac{(80\ln 2 - 18.5s)2^{7} + 18.5s}{\ln 2}$ $128\times 80\ln 2 - 127\times 18.5s = 0$ $s = \frac{128\times 80\ln 2}{127\times 18.5} = 3.02$ (to 3s.f.) The least number of stalks needed is 4.	

	Method 2: Given $h=18.5$. When $x=7$, $N\le 0$. $0\ge \dfrac{(80\ln 2-18.5s)2^7+18.5s}{\ln 2}$ $128\times 80\ln 2-127\times 18.5s\le 0$ $s\ge \dfrac{128\times 80\ln 2}{127\times 18.5}=3.02$ (to 3s.f.) The least number of stalks needed is 4.	

Source: 2023 Y6 CT Q6		
7	The plane π_1 contains the point $(2,-2,1)$ and the line $\mathbf{r}=\begin{pmatrix}0\\1\\8\end{pmatrix}+\lambda\begin{pmatrix}0\\2\\3\end{pmatrix}$, where λ is a parameter.	
	(a)	Find an equation of π_1 in scalar product form. [2]
	The plane π_2 has vector equation $\mathbf{r}=a\begin{pmatrix}0\\-1\\4\end{pmatrix}+b\begin{pmatrix}1\\2\\-1\end{pmatrix}$, where a and b are parameters.	
	(b)	Find the acute angle between π_1 and π_2. [3]
	The line l passes through the point A with position vector $4\mathbf{i}+5\mathbf{j}+6\mathbf{k}$ and is parallel to both π_1 and π_2.	
	(c)	Find a vector equation for l. [2]
	The plane π_3 with the cartesian equation $3x+2y+mz=0$ is also parallel to l.	
	(d)	Show that $m=-6$. [1]
	(e)	Find $\overrightarrow{OA}\cdot(3\mathbf{i}+2\mathbf{j}-6\mathbf{k})$. [1]
	(f)	Find the distance between l and π_3. [2]

	(g)	Hence or otherwise, find an equation of the line which is a reflection of l in π_3. [2]

Solution

(a) [2]

A normal vector to $\pi_1 = \left(\begin{pmatrix}0\\1\\8\end{pmatrix}-\begin{pmatrix}2\\-2\\1\end{pmatrix}\right)\times\begin{pmatrix}0\\2\\3\end{pmatrix}=\begin{pmatrix}-2\\3\\7\end{pmatrix}\times\begin{pmatrix}0\\2\\3\end{pmatrix}=\begin{pmatrix}-5\\6\\-4\end{pmatrix}$

π_1 : $\mathbf{r}\cdot\begin{pmatrix}-5\\6\\-4\end{pmatrix}=\begin{pmatrix}2\\-2\\1\end{pmatrix}\cdot\begin{pmatrix}-5\\6\\-4\end{pmatrix}=-10-12-4=-26$

(b) [3]

A normal vector to $\pi_2 = \begin{pmatrix}0\\-1\\4\end{pmatrix}\times\begin{pmatrix}1\\2\\-1\end{pmatrix}=\begin{pmatrix}-7\\4\\1\end{pmatrix}$

Let the acute angle between π_1 and π_2 be θ.

$$\cos\theta=\frac{\left|\begin{pmatrix}-5\\6\\-4\end{pmatrix}.\begin{pmatrix}-7\\4\\1\end{pmatrix}\right|}{\sqrt{(-5)^2+6^2+(-4)^2}\sqrt{(-7)^2+4^2+1^2}}=\frac{55}{\sqrt{77}\sqrt{66}}$$

$\theta=39.5°$ (to 1 d.p.)

(c) [2]

A vector parallel to both π_1 and π_2 is perpendicular to both normal vectors, one such vector will be:

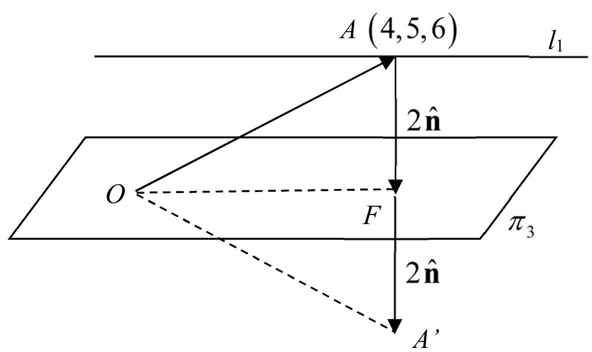
$$\begin{pmatrix}-5\\6\\-4\end{pmatrix}\times\begin{pmatrix}-7\\4\\1\end{pmatrix}=\begin{pmatrix}6+16\\28+5\\-20+42\end{pmatrix}=\begin{pmatrix}22\\33\\22\end{pmatrix}//\begin{pmatrix}2\\3\\2\end{pmatrix}$$

Therefore l: $\mathbf{r}=\begin{pmatrix}4\\5\\6\end{pmatrix}+\beta\begin{pmatrix}2\\3\\2\end{pmatrix}$, where $\beta\in\mathbb{R}$.

__**Alternative Method**__

$$\begin{aligned}\pi_1 &: \ -5x+6y-4z=-26\\ \pi_2 &: \ -7x+4y+z=0\end{aligned} \Rightarrow \begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}-\frac{52}{11}+z\\-\frac{91}{11}+\frac{3}{2}z\\z\end{pmatrix} \text{ by GC}$$

Therefore l: $\mathbf{r}=\begin{pmatrix}4\\5\\6\end{pmatrix}+\beta\begin{pmatrix}1\\\frac{3}{2}\\1\end{pmatrix}$, where $\beta\in\mathbb{R}$.

(d) **[1]**	l is parallel to π_3, therefore the direction vector of l is perpendicular to the normal of π_3 $\begin{pmatrix}2\\3\\2\end{pmatrix}.\begin{pmatrix}3\\2\\m\end{pmatrix}=0\Rightarrow 6+6+2m=0\Rightarrow m=-6$ (shown)
(e) **[1]**	$\overrightarrow{OA}\cdot\begin{pmatrix}3\\2\\-6\end{pmatrix}=\begin{pmatrix}4\\5\\6\end{pmatrix}\cdot\begin{pmatrix}3\\2\\-6\end{pmatrix}=-14$
(f) **[2]**	Since l is parallel to π_3, the distance between l and π_3 is the same as the perpendicular distance from any point on l to π_3. Observe that point A is on l and the origin is on π_3. The distance between l and π_3 =length of projection of $\overrightarrow{OA}$ on $\begin{pmatrix}3\\2\\-6\end{pmatrix}$ $=\dfrac{\left\lvert\begin{pmatrix}4\\5\\6\end{pmatrix}\cdot\begin{pmatrix}3\\2\\-6\end{pmatrix}\right\rvert}{\sqrt{3^2+2^2+(-6)^2}}=\dfrac{14}{7}=2$ units
(g) **[2]**	**Hence Method 1:** From part (e), note that the angle between $\overrightarrow{OA}$ and $\mathbf{n}=3\mathbf{i}+2\mathbf{j}-6\mathbf{k}$ is an obtuse angle because the dot product is -14 which is a negative number. Therefore $\overrightarrow{OA'}=\overrightarrow{OA}+4\hat{\mathbf{n}}$ as illustrated in the diagram below.  where point F is the foot of perpendicular from the point $A(4,5,6)$ to π_3 and point A' is the point of reflection of A in π_3. Hence, $\overrightarrow{OA'}=\begin{pmatrix}4\\5\\6\end{pmatrix}+2\times2\times\frac{1}{7}\begin{pmatrix}3\\2\\-6\end{pmatrix}=\frac{1}{7}\begin{pmatrix}40\\43\\18\end{pmatrix}$

	The reflection of l in π_3 is $\mathbf{r}=\frac{1}{7}\begin{pmatrix}40\\43\\18\end{pmatrix}+\gamma\begin{pmatrix}2\\3\\2\end{pmatrix}$, where $\gamma\in\mathbb{R}$.
	Hence Method 2: Check that the point with position vector $\begin{pmatrix}4\\5\\6\end{pmatrix}+2\times\frac{1}{7}\begin{pmatrix}3\\2\\-6\end{pmatrix}=\frac{1}{7}\begin{pmatrix}34\\39\\30\end{pmatrix}$ lies on the plane π_3, so $\begin{pmatrix}3\\2\\-6\end{pmatrix}$ is in the same direction as $\overrightarrow{AF}$. Therefore $\overrightarrow{OA'}=\overrightarrow{OA}+2\overrightarrow{AF}=\begin{pmatrix}4\\5\\6\end{pmatrix}+2\times2\times\frac{1}{7}\begin{pmatrix}3\\2\\-6\end{pmatrix}=\frac{1}{7}\begin{pmatrix}40\\43\\18\end{pmatrix}$ The reflection of l in π_3 is $\mathbf{r}=\frac{1}{7}\begin{pmatrix}40\\43\\18\end{pmatrix}+\gamma\begin{pmatrix}2\\3\\2\end{pmatrix}$, where $\gamma\in\mathbb{R}$.
	Otherwise Method (not the most efficient way in this context): Let F be the foot of perpendicular from the point $A(4,5,6)$ to π_3 and let A' be the reflection point of point A in π_3. Then $\overrightarrow{OF}=\begin{pmatrix}4\\5\\6\end{pmatrix}+p\begin{pmatrix}3\\2\\-6\end{pmatrix}$ for some $p\in\mathbb{R}$, sub into π_3: $\left(\begin{pmatrix}4\\5\\6\end{pmatrix}+p\begin{pmatrix}3\\2\\-6\end{pmatrix}\right)\cdot\begin{pmatrix}3\\2\\-6\end{pmatrix}=0\Rightarrow -14+49p=0\Rightarrow p=\frac{2}{7}$ $\overrightarrow{OF}=\begin{pmatrix}4\\5\\6\end{pmatrix}+\frac{2}{7}\begin{pmatrix}3\\2\\-6\end{pmatrix}=\frac{1}{7}\begin{pmatrix}34\\39\\30\end{pmatrix}$ By ratio theorem, $\overrightarrow{OF}=\frac{\overrightarrow{OA}+\overrightarrow{OA'}}{2}\Rightarrow\overrightarrow{OA'}=2\overrightarrow{OF}-\overrightarrow{OA}=\frac{2}{7}\begin{pmatrix}34\\39\\30\end{pmatrix}-\begin{pmatrix}4\\5\\6\end{pmatrix}=\frac{1}{7}\begin{pmatrix}40\\43\\18\end{pmatrix}$ The reflection of l in π_3 is $\mathbf{r}=\frac{1}{7}\begin{pmatrix}40\\43\\18\end{pmatrix}+\gamma\begin{pmatrix}2\\3\\2\end{pmatrix}$, where $\gamma\in\mathbb{R}$.

Source: 2023 Y6CT Q7	
8	A firm claims that, on average, their workers spend no more than 8 hours a day at the construction site. The amount of time clocked, x hours, by each worker from a random sample of 150 workers, is summarised as follows. $$\sum x = 1305\,,\quad \sum x^2 = 12751\,.$$ Test, at the 1% significance level, whether the firm's claim is valid. [5]
Solution	
[5]	Unbiased estimate of population mean, $\bar{x} = \dfrac{1305}{150} = 8.7$ hours Unbiased estimate of population variance $= s^2$ $= \dfrac{1}{149}\left[\sum x^2 - \dfrac{\left(\sum x\right)^2}{150}\right]$ $= \dfrac{1}{149}\left[12751 - \dfrac{(1305)^2}{150}\right]$ $= \dfrac{2795}{298}$ $= 9.3792 \text{ hours}^2 \,(5\text{s.f})$ Let X hours be the time spent by each worker at the site daily and μ hours be the population mean time spent by each worker at the site daily. Null Hypothesis $\mathrm{H}_0 : \mu = 8$ Alternative Hypothesis $\mathrm{H}_1 : \mu > 8$ Perform a 1-tail test at 1% significance level. Under H_0, since sample size 150 is large, by Central Limit Theorem, $\overline{X} \sim \mathrm{N}\left(8, \dfrac{2795}{298(150)}\right)$ approximately. From the sample, $\bar{x} = 8.7$ Using a z-test, p-value $= \mathrm{P}\left(\overline{X} \geq 8.7\right) = 0.00256 < 0.01$ Since p-value = 0.00256 < 0.01, we reject H_0 and conclude that there is sufficient evidence, at the 1% significance level, that the firm's claim is not valid.

Source: 2023 Y6CT Q8

9 Colour Vision Deficiency is the decreased ability to see differences in colours. Someone with normal colour vision will be able to see 6 colours (Red, Orange, Yellow, Green, Blue, Violet) as 6 bands of colours.

Deutan, who has colour vision deficiency, is not able to differentiate the following two pairs of colours:

(I) Red and Green,
(II) Blue and Violet.

For 6 colours (Red, Orange, Yellow, Green, Blue, Violet) of a particular shade, when the above pairs of colours are placed next to each other, it will look like a single band of colour to Deutan. Some examples of his vision of these 6 colours are as follows.

<u>Arrangement</u> of these 6 colours in a line	**Number of bands it will appear to Deutan**
Orange, Yellow, Red, Green, Blue, Violet	4
Red, Orange, Yellow, Green, Violet, Blue	5
Violet, Red, Orange, Yellow, Green, Blue	6

Find the number of ways these 6 colours can be arranged such that Deutan will see exactly

(a) 4 bands of colour, [2]

(b) 5 bands of colour, [3]

(c) 6 bands of colour. [2]

Solution

(a) [2]

To have only 4 colour bands, red/green must be together and blue/violet must be together.

Red and Green | Blue and Violet | Orange | Yellow

So we have 4 objects to arrange, 2 of which are groups of 2. So the total number of ways $= 4!\times 2!\times 2! = 96$.

(b) [3]

To have 5 colour bands, one of the pairs (red/green or blue/violet) must be separated, and the other pair together.

Consider red/green together.

<u>Method 1 (Grouping followed by Slotting)</u>

↓ Red and Green ↓ Orange ↓ Yellow ↓

Step 1 : arrange Orange, Yellow, Red/Green

Number of ways =3!2!

Step 2 : insert blue and violet in between Orange, Yellow, Red/Green

Number of ways = $\binom{4}{2}\times 2!$ or ${}^{4}P_{2}$

Total number of ways (after considering Blue/Violet as the grouped item in Step 1) =

$$\left[3!2!\binom{4}{2}\times 2!\right]2 = 288$$

Method 2

Step 1 : arrange Orange, Yellow, Blue, Violet, Red/Green

Red and Green	Blue	Violet	Orange	Yellow

Number of ways = $2!5!$

Step 2 : Consider those cases with Blue and Violet together also

Red and Green	Blue and Violet	Orange	Yellow

Number of ways = 96 (from part (i))

Total number of ways (after considering Blue/Violet as the grouped item in Step 1) =
$2[2!5! - 96] = 288$

(c) **[2]**

Since Duetan will see either 4, 5 or 6 colour bands,

required number of ways = 6! – (i)'s answer – (ii)'s answer = 720 – 96 – 288 = 336

Alternatively [Consider Cases]

Case 1: arrange O/Y, then slot in R/G, followed by B/V

Number of ways $= 2!\times\binom{3}{2}2!\times\binom{5}{2}2! = 240$

Note that the cases with RBG, RVG, GBR, GVR are not included in this case.

Case 2 : either one of RBG, RVG, GBR, GVR is a group

For each of the 4 cases, there will be a lone V, B, V and B, respectively, and we need to arrange the group, the lone V or B, and the other two colours O and Y, so therefore 4!.

Number of ways = $4!\times 4 = 96$

Total number of ways = 240 + 96 = 336

Source: 2023 Y6CT Q9

10 Two families plan to go on a hiking trip together. There are 10 people from the Lee family, consisting of 3 married couples, 3 single men and 1 single woman. There are 14 people from the Tan family, consisting of 4 married couples, 1 single man and 5 single women. Two people are randomly chosen to organise the hiking trip.

(a) Show that the probability that they are both from the Lee family is $\frac{15}{92}$. [1]

(b) Find the probability that they are a man and woman from the same family. [3]

(c) Find the probability that they are married to each other given that they are a man and a woman from the same family. [3]

Solution

(a) [1]

P(both are from the Lee family)

$$=\frac{{}^{10}C_2}{{}^{24}C_2}=\frac{45}{276}=\frac{15}{92} \text{ (shown)}$$

OR : $\frac{10}{24}\times\frac{9}{23}$

(b) [3]

P(man and woman from the same family)

=P(man and woman from the Lee family)

+ P(man and woman from the Tan family)

$$=\frac{{}^{6}C_1\,{}^{4}C_1}{{}^{24}C_2}+\frac{{}^{5}C_1\,{}^{9}C_1}{{}^{24}C_2}$$

$$=\frac{1}{4}$$

OR : $\left(\frac{6}{24}\times\frac{4}{23}\right)2!+\left(\frac{5}{24}\times\frac{9}{23}\right)2!$

(c) [3]

$\mathrm{P}\left(\text{married to each other}\mid\text{man and a woman from the same family}\right)$

$$=\frac{\text{no. of ways to choose a couple}}{\text{no. of ways to choose a man and woman from the same family}}$$

$$=\frac{{}^{3}C_1+{}^{4}C_1}{{}^{6}C_1\,{}^{4}C_1+{}^{5}C_1\,{}^{9}C_1}$$

$$=\frac{7}{69}$$

OR

A : the two chosen people are a married couple

B : man and woman from the same family

$$\mathrm{P}\left(A \mid B\right)=\frac{\mathrm{P}\left(A\cap B\right)}{\mathrm{P}\left(B\right)}$$

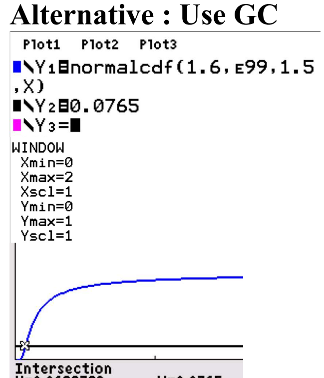
$$=\frac{\mathrm{P}\left(\text{married couple from the Lee family}\right)+\mathrm{P}\left(\text{married couple from the Tan family}\right)}{\mathrm{P}\left(B\right)}$$

$$=\frac{\left(\frac{6}{24}\times\frac{1}{23}\right)+\left(\frac{8}{24}\times\frac{1}{23}\right)}{\frac{1}{4}}$$

$$=\frac{7}{69}$$

Source: 2023 Y6CT Q10

11	In this question you should state clearly the parameters of any normal distributions you use. The masses in kilograms of chickens have the distribution $\mathrm{N}(1.5,\sigma^2)$ such that 7.65% of them have a mass greater than 1.6 kg.		
	(a)	Find the value of σ .	[2]
	The masses in kilograms of ducks have the distribution $\mathrm{N}(2.0, 0.08^2)$.		
	(b)	Find the probability that the mass of a randomly chosen chicken is less than 1.6 kg and the mass of a randomly chosen duck is more than 1.9 kg.	[2]
	(c)	Chickens and ducks are sold at \$11 per kg and \$18 per kg respectively. Find the probability that the total selling price of 4 randomly chosen chickens differs from twice the selling price of a randomly chosen duck by at least \$5.	[4]

Solution		**Comments**
(a) **[2]**	Let X be the mass in kg of a chicken. Then $X \sim \mathrm{N}\left(1.5,\sigma^2\right)$ $\mathrm{P}\left(X>1.6\right)=0.0765$ $\mathrm{P}\left(Z>\frac{1.6-1.5}{\sigma}\right)=0.0765$ From GC, $\frac{0.1}{\sigma}=1.4290$ $\sigma=0.069979$ (5 s.f.) $\sigma=0.0700$ (3 s.f.)	**Alternative : Use GC** 

(b) **[2]**	Let Y be the mass in kg of a duck. Then $Y \sim \mathrm{N}\left(2.0, 0.08^2\right)$ Required probability $= \mathrm{P}(X < 1.6)\mathrm{P}(Y > 1.9)$ $\approx (1 - 0.0765)(0.89435)$ $= 0.826$ (3 s.f)	
(c) **[4]**	Let $S = 11(X_1 + X_2 + X_3 + X_4) - 18(2Y)$ $\mathrm{E}(S) = 11(4)\mathrm{E}(X) - 18(2)\mathrm{E}(Y) = 44(1.5) - 36(2) = -6$ $\mathrm{Var}(S) = 11^2(4)\mathrm{Var}(X) + 18^2(2)^2\,\mathrm{E}(Y)$ $= 11^2(4)0.069979^2 + 18^2(2)^2\,0.08^2 = 10.665$ Then $S \sim \mathrm{N}(-6, 10.665)$ $\mathrm{P}(\lvert S\rvert \geq 5) = 1 - \mathrm{P}(-5 \leq S \leq 5) = 0.621$ (3 s.f.)	Since (a) was not a 'Show' question, we should use 5 s.f. or higher for σ in the calculation of $\mathrm{Var}(S)$.

Source: 2023 Y6CT Q11		
12	By the end of 2022, the percentages of a population who has received V dose(s) of Covid-19 vaccine are shown in the table below, with no one receiving 5 or more doses of vaccine.	
	(a)	Given that $\mathrm{E}(V) = 2.72$, find $\mathrm{Var}(V)$. [4]
	Mr Lee is doing a survey on people's perspectives on the Covid-19 vaccine.	
	(b)	Find the probability that the total number of doses of vaccine received by 3 randomly selected surveyees is more than 10. [2]
	Moderna's and Pfizer-BioNTech's bivalent vaccine are available at a particular vaccination centre. On average 60% of walk-ins opt for Moderna's bivalent vaccine at this centre. Let M denote the number of people, out of n walk-ins surveyed by Mr Lee on a particular day, that choose to receive the Moderna's bivalent vaccine.	
	(c)	Explain, in the context of this question, the assumptions needed to model M by a binomial distribution. [2]
	Assume now that these assumptions do in fact hold.	

v	0	1	2	3	4
Percentage of population	7	a	b	69	9

(d)	Find $\mathrm{P}(4 \le M < 8)$ when $n = 10$.	[2]
(e)	Find the least number of people Mr Lee needs to survey to have at least a 90% chance of at least 15 surveyees choosing to receive Moderna's bivalent vaccine.	[3]

	Solutions	**Comments**
(a) **[4]**	$7+a+b+69+9=100 \Rightarrow a+b=15$ $\mathrm{E}(V)=0.01a+0.02b+3\times 0.69+4\times 0.09=2.72$ $\Rightarrow 0.01a+0.02b=0.29$ By GC, $\begin{aligned} a+\quad b&=15 \\ 0.01a+0.02b&=0.29 \end{aligned} \Rightarrow \begin{aligned} a&=1 \\ b&=14 \end{aligned}$ $\mathrm{E}(V^2)=1^2(0.01)+2^2(0.14)+3^2(0.69)+4^2(0.09)=8.22$ $\mathrm{Var}(V)=\mathrm{E}(V^2)-[\mathrm{E}(V)]^2=8.22-2.72^2=0.8216$	Note that "percentage" instead of "probability" was used in the table.
(b) **[2]**	Let S be the total number of doses of vaccine received by 3 surveyees. $\mathrm{P}(S>10)=\mathrm{P}(S=11)+\mathrm{P}(S=12)$ $=\frac{3!}{2!}\mathrm{P}(V=3)[\mathrm{P}(V=4)]^2+[\mathrm{P}(V=4)]^3$ $=3\times 0.69\times 0.09^2+0.09^3$ $=0.017496$ (exact)	
(c) **[2]**	The event that one person chooses to receive the Moderna's bivalent vaccine is independent of the event that another person chooses to receive the Moderna's bivalent vaccine. The probability of a person choosing to receive the Moderna's bivalent vaccine remains constant at 0.6 for all walk-ins.	
(d) **[2]**	$M \sim \mathrm{B}(10, 0.6)$ when $n=10$ $\mathrm{P}(4\le M<8)=\mathrm{P}(M\le 7)-\mathrm{P}(M\le 3)$ $\approx 0.83271-0.054762$ $=0.778$ (3s.f.)	
	Alternatively, $\mathrm{P}(4\le M<8)=\sum_{m=4}^{7}\mathrm{P}(M=m)=0.778$ (3s.f.)	
(e) **[3]**	$M \sim \mathrm{B}(n, 0.6)$ $\mathrm{P}(M\ge 15)\ge 0.9$ $\mathrm{P}(M\le 14)\le 0.1$ n = 29: $\mathrm{P}(M\le 14)$ = $0.1362>0.1$ n = 30: $\mathrm{P}(M\le 14)$ = $0.0971<0.1$ From GC, least n = 30	